

# Deep-hole rigidity of the Clebsch hexagon code

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## Abstract

We characterize the exceptional non-GRS  $[6, 3, 4]_{11}$  MDS code by the geometry of its maximum-distance syndromes. Among six-arcs of  $\text{PG}(2, 11)$ , the projective deep-hole syndrome locus lies on a conic exactly for the Clebsch class. The syndrome conic gives an exact decoder, recovers the automorphism group, and determines an invariant bipartition of the ambiguous leader supports. We also give the sharp nearest-conic gap and classify the full-conic extension-locus examples for arcs of size 4 through 7.

**Keywords.** Clebsch hexagon; MDS code; deep hole; finite projective plane; arc; conic; decoding.

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## 1 Introduction

The principal question is whether the deep-hole syndrome geometry recognizes the Clebsch hexagon without an a priori symmetry assumption. The main theorem answers this question and organizes the decoding, reconstruction, and classification consequences.

**Transfer gate.** C575 will import the exact theorem statements, hypotheses, citations, and proof-mode declarations from the pinned focused source. This file records their narrative order but does not yet replace that source.

## 2 The code–arc dictionary

Fix the projective and coding conventions once, identify uncovered points with weight-three syndrome directions, and isolate the extension locus used throughout the paper.

## 3 The Clebsch hexagon and its code

Give the six points, the explicit parity-check matrix, the automorphism action, and the nonsingular conic containing the twelve distinguished syndrome directions.

## 4 Syndrome geometry, decoding, and automorphisms

Prove that the syndrome conic is a constant-time distance oracle. Then present the complete fifteen-class ambiguity census and derive the automorphism consequences without interrupting the proof with replay details.

## 5 Rigidity and quantitative gaps

Separate the conceptual rigidity proof from the exhaustive finite calculation. State the qualitative characterization first, followed by the sharp nearest-conic gap and the low-degree strengthening.

## 6 Support bipartition and Brianchon reconstruction

Recover the Brianchon configuration from the decoder data and construct the intrinsic unordered  $10 + 10$  bipartition of ambiguous leader supports.

## 7 Why $q = 11$

Explain the arithmetic isolation of the Clebsch case, then prove the classification of full-conic extension loci for  $4 \leq k \leq 7$ . A short reflection-arrangement remark may remain only if it clarifies this classification without importing the factorization-memory machinery.

## 8 Verification architecture

Map each headline claim to its conventional proof, classical input, Lean theorem, certificate, or exact computation. Keep the exhaustive search boundary limited to the numerical gap and low-degree strengthening.

## Conclusion

Return to what the syndrome geometry reconstructs, why  $q = 11$  is isolated, and the open range  $k \geq 8$ .